



DSC 360 FINAL PROJECT RUBRIC [Note: Submit deliverables here. The project is due Friday, Dec. 5, but can be submitted without penalty up to the time of the final exam.]

Completion requirements

Due: Friday, December 5, 2025, 4:00 PM

DSC 360 Final Project

Building AI-Powered Applications Weight: 15% of course grade

The final project is a team-based mini-capstone. You will work in teams of **1–3** to **design, build, and evaluate** an LLM-based application and communicate your work in writing and in a short oral presentation.

We are looking for a **real, working LLM application, not just vibes.**

If your project starts to go off track at any point, please **contact me early** so that we can adjust scope and expectations together. Keeping your instructor in the loop is part of professional communication.

Deliverables

Use **this** Moodle assignment to submit your **final project deliverables**. Unless we have agreed on a different arrangement, package everything into a **single, well-organized project folder** with a short `README.txt` (or `README.md`) and upload that folder as a **.zip file**.

Your zipped project folder should include:

- **Code** (for example, as a complete project folder)
- **Data** needed to run your demo (small samples only if the full dataset is very large)
- **Configuration files** (for example, `.env.example`, TOML/JSON config, model settings)

- **Final written report** (Word or PDF format only)

Large files and datasets:

If your project involves **very large files or datasets** (for example, multi-GB logs, large media files, or a full raw corpus), **do not upload them directly to Moodle**.

Instead, you must **contact me in advance** so we can agree on an alternative (for example, a GitHub repository link, external storage link, or similar). Any external repository or folder should clearly reflect the state of the project as of the deadline.

If you are unsure whether something is “too big,” please ask before you upload.

See separate Moodle assignments to submit:

- Project proposal
- Milestone and progress reports, and partner and team feedback
- Final presentation slides

Important dates

- **Project work window:** Project work officially begins **Friday, November 14** and continues through the last two weeks of class.
- **Milestone plan check-in: Monday, November 17 at 10:20 AM** (one per team, see separate Moodle item).
- **Progress check-in: Monday, December 1 at 10:20 AM** (one per team, see separate Moodle item).
- **All project deliverables (except slides and progress reports) due: Friday, December 5 at 4:00 PM** (Code, data, configuration files, and final written report.)
- **Presentation slides due: Wednesday, December 10 at 1:00 PM** (PowerPoint or PDF uploaded to Moodle. No cloud links.)
- **Final presentations: Wednesday, December 10, 1:30–4:30 PM** (registrar final exam period).
- **Partner and team contribution report: Wednesday, December 10 by 4:30 PM** (Individual short report on team contributions; see section at the end.)

Use of LLMs and coding assistants

Full use of LLMs and coding assistants (for example, OpenAI, Claude, GitHub Copilot, Replit, and similar tools) is **allowed** for this project, as long as:

- You **document clearly in your written report** how you used these tools.
- You **understand** all code, prompts, and configurations that you submit.
- You remain in control of the project scope and design.

These tools often suggest ambitious or overly complex solutions. It is your responsibility to **steer the project**, keep the scope achievable, and focus on a system that you and your teammates can explain and maintain.

How the project will be graded

Your final project grade (0–100) is based on **five categories**:

1. **Design and Scope** – 25%
2. **Implementation and Functionality** – 35%
3. **Evaluation and Reflection** – 20%
4. **Written Communication (Reports and Check-Ins)** – 10%
5. **Oral Presentation** – 10%

Each category is assessed using four levels of achievement:

- **Excellent**
- **Solid**
- **Developing**
- **Weak**

The descriptions below explain what those levels look like for each category.

Important: Working application requirement

This course is about **building** AI-powered applications.

To earn **above 30 out of 100** on the final project:

- Your team must demonstrate a **real, working LLM application**.
- Demonstration can be a live demo during the presentation or a clearly recorded demo combined with reproducible code and instructions.

If there is **nothing to demonstrate** (that is, no working end-to-end run of your application), the **maximum** project score is **30 out of 100**, even if other components (design, writing, or slides) are stronger. Within that cap, partial credit is still possible.

If you are at risk of not having a working system, contact me as early as possible so that we can adjust the scope and expectations.

1. Design and Scope (25%)

What we are looking for: A clear, well-reasoned design for your LLM-powered application, with appropriate scope for a 1–3 person team over the project period.

Excellent: The problem is clearly defined and appropriately sized. The overall system design is coherent and well-motivated: you explain why you chose particular models, data sources, retrieval or RAG approaches, prompts, and safety measures. An architecture diagram or equivalent explanation makes it easy to see how the pieces fit together.

Solid: The design is mostly clear and appropriate, with only minor gaps or fuzzy areas. The choices of model, data, and approach are reasonable for the problem, even if not fully justified in detail.

Developing: The scope is somewhat unfocused or poorly matched (for example, clearly too large for the time available, or too small to be interesting). The design is hard to follow or has obvious missing elements about how the system is supposed to work.

Weak: The problem is poorly specified or constantly shifting. There is little evidence of deliberate design beyond “we tried some prompts and saw what happened.”

2. Implementation and Functionality (35%)

What we are looking for: A **real, working** LLM application that runs end-to-end and demonstrates meaningful functionality.

Excellent: The system runs end-to-end with minimal manual intervention. Core components (for example, retrieval, calling the model, processing the output, and any safety checks) are implemented and integrated. The code is reasonably organized, configuration is separated from logic (for example, model names, keys, and paths are not hard-coded all over), and it is realistically reproducible on a lab machine.

Solid: The system runs on typical examples, but may be somewhat brittle or require simple manual steps (for example, editing a configuration file or restarting a script). Most core components are present and working together. The application clearly corresponds to the design you described.

Developing: Major parts of the system are missing or only partially implemented. It is difficult to obtain a full end-to-end run without significant instructor intervention. You can see the general idea, but it is not really usable.

Weak: There is no meaningful end-to-end run of the application. Only isolated code fragments, experiments, or descriptions are present, with nothing that clearly counts as a working system.

3. Evaluation and Reflection (20%)

What we are looking for: Evidence that you tested your system, collected results, and thought about what they mean.

Excellent: The team ran a meaningful evaluation (for example, 15–30 test prompts or tasks, or a carefully curated test set) and recorded the results clearly. The project compares at least a few variants (different prompts, models, parameters, or retrieval strategies) or reflects on design tradeoffs. The written report and presentation include an honest discussion of strengths, weaknesses, and failure modes, and explain why you ended up with the design you chose.

Solid: The team ran some tests and documented results, but the evaluation is more modest or ad hoc. There is some reflection on what works well and what does not, but the discussion is not very deep.

Developing: There is very little systematic evaluation. Claims about performance are mostly anecdotal, with few concrete examples or metrics. Limitations and failure cases receive minimal attention.

Weak: There is almost no evidence of testing or reflection. The project simply claims that the system works (or does not) without showing how the team knows.

4. Written Communication (Reports and Check-Ins) (10%)

This category covers:

- The **final written report** (main component).
- The **project proposal**, milestone plan, progress check-ins, and required brief peer or partner evaluations.

Progress reports and short check-ins are graded for **completion**, but missing them affects this communication score.

Timeliness matters in this category.

Excellent: The final written report is clear, polished, and well-structured, with sensible headings, readable layout, and helpful visuals (for example, diagrams, tables, or

screenshots). Grammar and style support clarity. **All required written deliverables**, including the proposal, milestone plan, progress reports, and peer evaluations, are submitted on time and show genuine effort.

Solid: The final report is generally clear and well-organized, with only minor issues in writing, layout, or visuals. All required written deliverables are submitted, but one may be slightly late or less detailed than ideal.

Developing: The final report is harder to follow, with organizational issues, missing sections, or frequent writing problems. One or more progress reports or other required written components are missing or very minimal.

Weak: The final written report is extremely brief, incomplete, or missing. Several required written deliverables are missing. If one or more progress reports are missing, this category will fall in the Weak range. If the final report is missing entirely, this category will effectively receive no credit.

5. Oral Presentation (10%)

- **Time:** 10–12 minutes per team.
- **Slides:** PowerPoint or PDF only. No links to any cloud service.
 - Slides must be uploaded to Moodle by **1:00 PM on the presentation day**.

What we are looking for: Clear oral communication of your problem, your design choices, and what your system can actually do, supported by well-prepared visuals and a meaningful demonstration.

Audiovisual expectations:

- Slides should be clean, readable, and visually appropriate (font size, contrast, and layout).
- Slides should support and clarify your talk, not distract from it.
- A live demo is strongly preferred if your system interacts with users in real time.
 - If your project focuses on batch data extraction or similar, a recorded demo or a careful walkthrough with screenshots is acceptable (submitted separately).
- The presentation should show clear evidence of skillful preparation and careful rehearsal.

Excellent: A polished, well-prepared presentation that stays within the time limit. Slides are clear and professional and enhance understanding. The demo (live or recorded)

effectively shows the system working as described. Delivery shows that the team has practiced and thought about how to explain the project to an audience.

Solid: A generally clear presentation with minor issues in pacing, organization, or visuals. The demo works but may be slightly clunky or limited. Slides are mostly effective but could be improved.

Developing: The presentation is hard to follow, disorganized, or poorly timed. Slides are cluttered, hard to read, or incomplete. The demo is partial or does not clearly show what the system can do.

Weak: The presentation is incomplete or very confusing. Slides are missing or unusable. There is no meaningful demonstration of the system.

Partner and team contribution report

By **Wednesday, December 10 at 4:30 PM**, each student will submit an individual, short **partner and team contribution report** on Moodle. This report will briefly describe:

- How responsibilities were divided in your team,
- The level of contribution and engagement from each teammate, and
- Any significant issues that arose.

I will take these reports into consideration when assigning final project grades. In some cases, teammates may receive **different individual project scores** based on their level of contribution.

If you have any questions about this rubric, about the expectations, or about whether your project is on a reasonable path, please talk with me. The earlier you reach out, the more I can help.